DATA ITEM DESCRIPTION
Title: Nuclear Hardness and Survivability (NH&S) Program Plan

Number: DI-ENVR-82097
AMSC Number: F9759
DTIC Applicable:
Preparing Activity: AF-27
Applicable Forms: None

Approval Date: 20161209
Limitation:
GIDEP Application: No
Project Number: ENVR-2016-005

Use/Relationship: This Data Item Description (DID) contains the format and content requirements for the NH&S Program Plan. This data is produced as a result of tasking in 5.3.1.2 and 5.3.1.2.1 of MIL-STD-1766C but because it is a plan for execution of all tasks in that standard, its scope is not restricted to these two paragraphs. It is initially due at PDR but needs to be required by contract at other points before (e.g., during TMRR phase) or after.

This DID is applicable to system development contracts during the engineering and manufacturing development and production phases, and to other contracts when an NH&S program is required. It contains the format and content requirements for the NH&S Program Plan that is needed to organize and conduct the activities and tasks defined in MIL-STD-1766 (latest revision). The essential content of DIDs DI-ENVR-80263 (hardness assurance plan), DI-ENVR-80264 (hardness maintenance plan), and DI-ENVR-80265 (hardness surveillance plan) has been incorporated into this DID through the use of updates to the NH&S Program Plan at the appropriate point in a program as defined in MIL-STD-1766 (latest revision).

This plan provides the basis for the procuring agency's determination that the contractor's NH&S program will meet contractual requirements in a cost-effective manner and is used by the procuring agency to review, evaluate, and approve all aspects of the contractor's NH&S program. It also serves as an ongoing means for establishing and assuring mutual understanding between the procuring agency and the contractor as to the scope and content of the contractor's NH&S activities.

The nature of the contract will be a factor in deciding the number of versions of the plan that need to be called for in the contract. Updates to the plan are needed at appropriate points in the program to include the Hardness Assurance planning, the Hardness Maintenance planning, and the Hardness Surveillance planning.

This DID describes a NH&S program description which is new and unique. Comments, suggestions, or questions on this document should be addressed to the Defense Standardization Program Office (DSPO), 1551 Wyoming Blvd. SE Kirtland AFB, NM 87117-5624 or emailed to afnwc.dspo.workflow@us.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.dla.mil.

Requirements:
1. **Format.** The plan shall be in the contractor's format except where format is specified herein and shall be consistent with the content and tasking of MIL-STD-1766 (latest revision).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:59Z
Check the source to verify that this is the current version before use.

Page size shall generally be 8 ½ x 11" except where necessary, 11 x 17" may be used as a foldout (folded to 8 ½ x 11").

2. **Content.** The report content shall be as follows with additions as the contractor deems necessary to describe the NH&S program.

3. **General content.** The NH&S Program Plan shall cover all aspects of the NH&S program, including the schedule for execution based on program milestones with sufficient details of specific tasks, analyses, and tests to allow for budgeting and scheduling of program resources, and for directing and monitoring technical progress. The plan shall cover the methods and techniques for incorporating nuclear survivability into design, development, and system integration, and for conducting a comprehensive NH&S validation and qualification program. The engineering organization responsible for NH&S shall be identified and the organizational structure in which it operates defined (e.g. organization chart and associated discussion) showing how all other organizations will interact to receive proper direction and communication that ensures cognizance throughout the program team of the NH&S tasking, methods, and purposes.

4. **Executive Summary**. An executive summary shall identify and briefly discuss the essentials of the program plan and identify in a list, the deliverable end product(s) (hardware and NH&S data). Include a cross index in the format shown below to identify the relationship between all contract program tasks, tasks planned to implement the contract tasks, and (1) applicable specifications or standards cited by the contract work statement, (2) other referenced documents, and (3) contractor policies and standards.

Applicable paragraph of contract tasking (e.g., SOW) or prime item development specification (PIDS) or applicable task designed to implement the contract/PIDS	Applicable paragraph of the NH&S program plan that discusses the details of that task	Other referenced documents related to the task and/or requirement, including computer codes to be employed	Company procedures and controls

Format for task and requirements cross-reference matrix

5. **Specific Content:** The program plan shall be prepared consistent with the tasking of MIL-STD-1766 (latest revision) and shall include the information as described below:

   a. Identification of the work, analyses and tests to be accomplished for each applicable task, and the time phasing involved, including a calendar schedule which details the technical and program management reviews to be held along with technical interchange and other program meetings.

   b. The plan shall detail the combination of analyses and test that will be employed to validate and qualify (MIL-STD-1766C , 5.3.5.1 and 5.3.5.2) the hardware being designed

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:59Z
Check the source to verify that this is the current version before use.

and/or produced. The plan shall identify the assembly level for tests and/or analyses to be performed and the relationships between requirements and the testing and analytical efforts. A summary of this part of the plan shall be provided in a nuclear environments and effect hardware element matrix (NEEHEM) as described in MIL-STD-1766C, 5.3.1.2 and in the example provided at the end of this DID. The text discussing the NEEHEM shall describe the rationale for and decision-making process employed to determine the mix of testing vs. analysis, identification test facilities and simulators to be used, test schedule, etc. When analysis is selected, the rationale shall demonstrate that the analysis is a technically adequate means to ensure compliance with nuclear environment performance requirements.

c. The plan shall include identification of computer codes and their intended application. An alternate version of the NEEHEM may be used.

d. Planned trade studies, including consideration of what will be done for hardness assurance, hardness maintenance and hardness surveillance tests and inspections planning and preparation.

e. The plan shall present a schedule of critical NH&S program milestones related to the overall program schedule and milestones. This time-phased schedule shall indicate start date, review date(s), and completion date for each key nuclear survivability task.

f. The plan shall present a general discussion of the integrated logistic support management approach to field a survivable system, including design, analysis, test, and documentation.

g. The plan shall delineate piece part, component, and subsystem tests to be performed for each specified nuclear environment and indicating relationships to analytical efforts. Details, such as sample size, test methods, test instrumentation; parameters to be characterized may be provided or reserved for separate delivery in the specific test plan for the item being tested.

h. The plan shall describe anticipated test facilities to be employed, anticipated schedules, and reporting. Details such as test configurations, test article orientation, exposure levels and test data to be obtained may be provided in the program plan or in the specific test plans for each item to be tested.

i. The plan shall describe the NH&S considerations that will be used as the basis for selection of piece parts, materials, device technologies, circuit and mechanical designs; The implementation plans for hardness critical items (HCI) and hardness critical processes (HCPs) shall be IAW MIL-STD-1766 (latest revision) and explained in the program plan, including the configuration control approach to control HCIs and HCPs, especially when non-developmental items are planned (MIL-STD-1766C, Table II and A.3.104).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:59Z
Check the source to verify that this is the current version before use.

j. If this contract will make use of any non-developmental or proprietary items, this plan shall detail how the nuclear survivability of these items will be evaluated, assured, maintained, and documented during all phases of life-cycle development, operation and support. The plan shall describe the method of control over contractor, subcontractor, and vendor NH&S programs that will ensure hardness requirements are met.

k. The plan shall describe the approach to the incorporation of NH&S issues and considerations into the appropriate life-cycle hardness (LCH, MIL-STD-1766C, 5.3.8.3 through 5.3.8.5.1) documentation as early as possible in the program IAW applicable MIL-STD-1766 (latest revision). The plan shall describe the procedures of MIL-STD-1766 (latest revision) that will be followed regarding the proper annotation of HCIs and HCPs in the Logistic Support Analysis and the Logistic Support Analysis Record when these items are required deliverables of the contract.

l. The plan shall delineate the aspects of the design that are anticipated to be driven by the nuclear survivability of the system, especially the anticipated need for custom or hardened parts, materials, or components and the basis of need for these items and delineate the design guidelines to provide standardized:

(1) Derating/design margins/safety factors in design;
(2) Design analysis techniques;
(3) Parts characterization/material properties data;
(4) Overall methodology for derating components, including the NH&S portion.

m. The plan shall identify the methodology for decision making relative to NH&S requirement trade-offs vs. other system requirements such as schedule, cost, mobility, reliability, producibility and integrated logistic support.

n. Nuclear Environments and Effects Hardware Element Matrix (NEEHEM). The program plan version of the NEEHEM is a matrix showing test and analysis plans for each major hardware division within the CI. The NEEHEM in this form is a planning tool for program scoping, scheduling, and status tracking. At PDR it shows the planned test facility and may contain samples size and other test information. An example NEEHEM is shown in the figure at the end of this DID. Other versions of the matrix are used in the NH&S DAR (under DI-ENVR-80266B) as a documentation organization tool to show all reports and/or other documents that fulfill the test/analysis intent of each box within the matrix.

6. **Hardness Assurance, Maintenance, and Surveillance content.** The plan shall discuss planning and preparation for hardness assurance (HA) (i.e., preservation of hardness throughout production) and hardness maintenance and surveillance (HM/HS) preparation (see MIL-STD-1766C, all content in section 5.3.8). The appropriate HA/HM/HS activities of MIL-STD-1766 (latest revision) shall be included regardless of the number of CIs or assembly level of hardware under contract. The program plan shall be prepared consistent with the information contained in MIL-STD-1766 (latest revision) and shall include the information as described below.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:59Z
Check the source to verify that this is the current version before use.

6.1 Hardness Assurance Content. At the appropriate point in the program as explained in MIL-STD-1766C (refer especially to 5.3.8.7 thru 5.3.8.10), the program plan shall be updated to include:

a. Identification of the hardware elements selected to undergo hardness assurance testing/ inspection. Include the rationale for each selection, including the rationale for the level of assembly chosen for test. For each selected item, identify the specific hardness related characteristics to be tested/inspected and the applicable nuclear environments, the sample size and test/inspection frequency recommended, and any special equipment and facility required to perform the test/inspection.

b. Identification of the procedural, documentation, and interface activities required of existing contractor control disciplines (or organizational entities) in support of hardness assurance. Include consideration of the following control disciplines: (1) configuration control; (2) parts/material control; (3) production/manufacturing control; and (4) quality assurance/quality control.

c. Description of the hardness assurance program management activity. Identify the approach that will be used to manage the hardness assurance program, and identify the responsibilities and interfaces of all contractor organizations/groups involved with hardness assurance planning and implementation, and the means by which their separate activities will be coordinated and integrated. Identify the organizational structure and means by which the progress and results of hardness assurance tests/inspections will be reviewed and evaluated in a timely manner, and corrective actions will be identified, implemented, and tracked in the event of validated problems or anomalies. Identify the management approach and procedures for the control and review of the data necessary to support and implement the hardness assurance activity.

6.2 Hardness Maintenance Content. At the appropriate point in the program as explained in MIL-STD-1766C (refer especially to 5.3.8.6, 5.3.8.7 and 5.3.8.16), the program plan shall be updated to include:

a. Description of the hardness data annotation activity. Identify all data items that will be hardness annotated in support of hardness maintenance and the procedures by which this annotation will be accomplished. Include a copy of any contractor-generated operating instruction, drafting manual amendment, or any other instructional document used by the contractor to implement the hardness annotation of engineering data. Identify the responsibilities and interfaces of all contractor organizations/groups involved with the planning and implementation of the hardness data activity and the means by which their separate activities will be coordinated and integrated.

b. Description of the hardness logistics support definition activity. Identify the procedures to be followed in accomplishing the incorporation of hardness requirements in the logistics support analysis (LSA) and the logistics support

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:59Z
Check the source to verify that this is the current version before use.

analysis records (LSAR). Identify guidelines/ground rules pertinent to the preservation of the hardness of the contractor's configuration item(s) (CI(s)) during repair/maintenance for use in accomplishing hardness annotation of the LSA/LSAR. Identify any hardness dedicated tests/inspections and associated equipment required during the repair/maintenance of the contractor's CIs for use in accomplishing hardness annotation of the LSA/LSAR and for preparations to transition equipment to the hardness maintenance implementation. Identify the resources required to implement such tests/inspections. Identify the responsibilities and interfaces of all contractor organizations/groups involved with the planning and implementation of the hardness logistics support definition activity and the means by which their separate activities will be coordinated and integrated.

6.3 Hardness Surveillance Content. At the appropriate point in the program as explained in MIL-STD-1766C (refer especially to 5.3.8.11, 5.3.8.13 and 5.3.8.15), the program plan shall be updated to include:

a. Identification of recommended hardness surveillance hardware candidates. Identify all hardware elements that are expected to undergo hardness degradations as a result of the conditions of operational life and the associated degradation mode(s) of concern for each item. Prioritize the list of candidate hardware elements in terms of the probability of occurrence during the operational life of the hardness degradation mode(s) postulated and the impact on the system mission should those degradations actually occur.

b. Identify how and when design considerations for hardness surveillance will be considered and the decision-making process to be used to determine whether such design features will be included. Address how the design feature can allow for easier implementation vs. the cost of other choices for carrying out the test/inspection and also the risk of introducing an additional vulnerability by the use of such a design feature. An example of such a design feature would be an antenna receiver placed in a location internal to the system that would allow for measurements of shielding attenuation of electromagnetic environments.

c. Identification of candidate tests/inspections which can be used to detect the hardness degradations identified. Optimize the candidate tests/Inspections with respect to the level of hardware assembly chosen for test/inspection and the minimization of Interference with routine field and depot operations. Address explicitly the possibilities for incorporating/ coordinating the candidate hardness surveillance tests/inspections with routine repair/maintenance activities and planned/existing aging and surveillance programs.

d. Identification of the contractor's recommendations for a hardness surveillance program for the hardware relevant to the contract. Include an estimate of the equipment, documentation, training, and other resources that would be required to implement the recommended program.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:59Z
Check the source to verify that this is the current version before use.

7. Management interface controls described in the plan:

a. Identification and definition of the interfaces between the NH&S program and closely related programs or elements within the contractor's organization. For the parts, materials, and processes standardization and control program, interface definition shall include those management procedures which will enable the contractor's NH&S organizational element to:

(1) Review and concur with hardness parameters and parameter values in specifications;
(2) Review and concur with the methods of control and audit of hardness critical manufacturing processes and controls;
(3) Review and concur with all Class I and Class II engineering changes affecting parts, materials, and processes and to review and concur with the decision to use a Class I or Class II process;
(4) Review and concur with hardness critical item (HCI) failure analysis;
(5) Serve as the ultimate authority within the contractor's organization relative to HCI designation and identification and hardness design feature selection and implementation with appropriate checks and balances to allow for cost and schedule considerations after appropriate management review to allow the best decision in the government's interest to be made; and
(6) Provide direct support to studies of the effects of operating life on HCIs.

Other NH&S program interfaces (if such functions exist within the contract scope) described shall include:

(1) Design engineering;
(2) Systems engineering;
(3) Parts control board;
(4) Configuration management;
(5) Reliability program;
(6) Maintainability program;
(7) Quality assurance program;
(8) Logistics engineering;
(9) Aging and surveillance program
(10) Worst case design analysis;
(11) System/cost effectiveness analysis;
(12) System acquisition life-cycle cost analysis;
(13) Assembly and checkout;
(14) Logistics support analysis;
(15) Operational test and evaluation program
(16) Technical Order HCI content, especially annotation for HCPs

b. Methodology and procedures by which the contractor shall ensure that the appropriate nuclear survivability program is executed in subcontracts shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:59Z
Check the source to verify that this is the current version before use.

described and the responsibilities division between the contractor and his subcontractors such that all NH&S tasks are either flowed-down to the subcontractor (where they are competent to perform) or assumed by the contractor for the subcontractor. The means by which any tasking to be performed by the contractor for a subcontractor will be coordinated between the contractor and subcontractor.

c. Actions to be taken to ensure nuclear survivability (to include validation, assurance and maintenance) for material to be procured elsewhere, including any non-development or proprietary items shall be delineated.

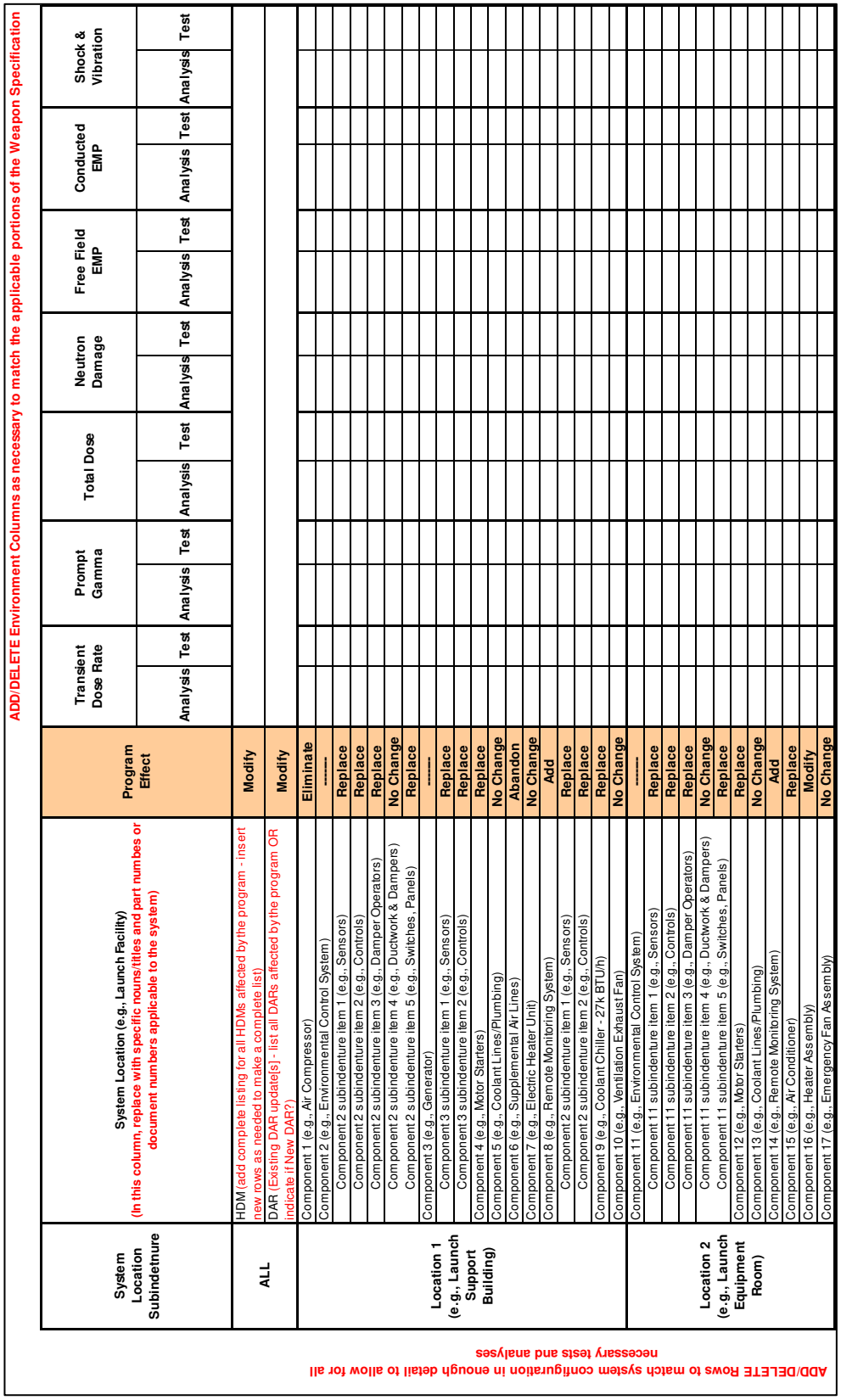
8. Program management described in the plan:

a. Responsibilities and authority of the NH&S Program Manager and the nuclear weapons effects experts assigned to the program. Identification of the personnel assigned to these functions with appropriate contact information.

b. Management process and actions required to ensure the successful application and accomplishment of NH&S requirements.

9. **Example NEEHEM:**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:59Z
Check the source to verify that this is the current version before use.



Example NEEHEM

10. End of DI-ENVR-82097

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:59Z
Check the source to verify that this is the current version before use.